



THE UNIVERSITY OF
MELBOURNE

CURRENT STUDENTS IA · FULL-ESTATE AUDIT

Improvements register

~75 UX, service, governance, content & technical improvements — grounded in the full 21,250-page link-graph scrape. June 2026.

University of Melbourne · June 2026

The University of Melbourne acknowledges the Traditional Owners of the unceded land on which we work, learn and live, and pays respect to their Elders, past and present, and to Indigenous Australians today.

Master deliverable for the digital team and DEAG. Every improvement is grounded in the full-scrape link-graph evidence pack (21,250 content pages, 20 domains, 261,652 contextual links after stripping 598 chrome patterns) and the existing analysis corpus. Severity + effort are tagged `[SEVERITY · effort]`. Citations point to `analysis/full-scrape/*.csv`, the raw `crawl/` corpus, and `analysis/*.md`.

Round-2 update. This register now carries ~75 improvements: the round-1 set (UX/IA, findability, seams, content, governance, accessibility) plus round-2 additions — Performance & Mobile (\$7), High-stakes Journeys (\$8), reproducible near-duplicate actions (\$9b) and content-freshness/ownership (\$9). Read the Scope & confidence section first.

Executive summary

The current-student journey is not served by one estate but by **20+ public systems with no shared information architecture or vocabulary** — and the link graph shows the sprawl is far deeper than the headline suggests: **415 distinct `*.unimelb.edu.au` hosts** appear in the contextual graph alone (Fine-Arts cohort-year microsites, one-lab-one-subdomain research sites, 19 single-purpose transaction `*.app` hosts), with no registry tying any host to an owner or a retirement date. Across the four faculty domains, current students are only a **~4–7% tenant** of estates that are built for reputation, research and recruitment; **no two faculties share a top-level taxonomy**, and the student entry is variously a top-level `/students`, a `/current-students`, or a section buried 2+ levels inside the prospective `/study` funnel. The two ends of the lifecycle are barely wired: after stripping nav chrome, **only ~97 contextual links reach the alumni estate** (it is nav-only, with no in-content graduation→alumni handoff) and **only ~100 reach central Careers Online** — the estate's two weakest seams. Meanwhile the canonical hub, `students.unimelb.edu.au` (833 pages, not the "74" the deck asserts), is itself **mid-migration**, dual-serving ~120 leaf pages under two parallel URL trees, and the estate's biggest single dependency — the Handbook (linked 3,380+ times, 7,819 contextual inbound) — sits on a different platform, governed to a visibly lower web standard. The opportunity is therefore overwhelmingly **structural and governance-led, not visual**: a thin standardised student overlay on the Squiz platform the University *already runs everywhere*, a subdomain/forms registry to stop re-fragmentation, completion of the in-flight hub migration, and a set of high-leverage template/redirect fixes that each clear thousands of links at once.

Important data caveat (carried from content-quality finding): the Handbook's "14% orphaned / thin" signal is substantially a **crawl artifact** — ~1,470 of 2,134 crawled handbook captures are Imperva/Incapsula bot-block shells (~887 bytes), and the 266 "orphans" are largely these blocked SPA pages. The *actionable* handbook issues (SPA with no server-rendered fallback, sub-baseline metadata, missing browsable indexes, no enrol action) stand on their own; the raw orphan/thinness percentages should be annotated as WAF/SPA artifacts and the handbook re-crawled before any "handbook is mostly thin" claim reaches the deck.

Scope, confidence & what is NOT yet verified

This register is grounded in the **public** estate scrape (21,250 content pages, 20 domains, the path-level link graph). Five honesty boundaries a reviewer must carry forward before any item ships:

- 1. The authenticated core is uncrawlable.** `my.unimelb` (SIS: enrolment, results, fees, timetable), Canvas/LMS, and the ~18 `*.app / forms.your` microsites sit behind login and appear only as outbound destinations. Every "wire X to my.unimelb / Careers Online / a form" recommendation is inferred from the public side — confirm the logged-in flow doesn't already do it (request screenshots / a walkthrough).
- 2. All 404 / redirect / "phantom-404" claims are URL-string inference, not live HTTP.** Nothing in this project issued an HTTP request. The `?a=` artifact (2.3), the dead-link sweep (2.4), the www-vs-bare pairs (5.4) and the parallel-tree leaves (1.1) need **HEAD-probe confirmation** of real status codes before remediation is scoped.
- 3. Findings are structural-evidence-only, not CX-validated** (carries `data-and-synthesis-issues.md` D6). The "add a bridge / CTA" items (3.1, 3.2, 8.x) rest on link-graph absence, not observed student failure — validate the costliest seams with real users.

4. **No analytics / traffic data.** Link-volume is the importance proxy throughout; it is not demand. The internal `in_c=` tracking params in the crawled URLs imply GA/search-log data exists — **requesting it is the single highest-value next step**, so priority can follow pages students actually hit.
5. **Two evidence notes:** near-duplicate claims are now reproducible via `analysis/full-scrape/near_dup.py` (4.x corrected below); the Handbook "14% orphaned" figure is largely an Incapsula bot-block artifact pending a JS-enabled re-crawl; and `students.unimelb` is **833 pages crawled / 679 with substantive content after de-dup** (both figures correct, different denominators — neither is the deck's "74").

Prioritised Top 12

#	Improvement	Category	Severity	Effort	Evidence (one line)
1	Finish the stalled hub migration & collapse the parallel URL trees on <code>students.unimelb</code>	UX & IA	HIGH	medium	120 leaf pages dual-served under <code>/your-course + /course-admin</code> and <code>/student-support + /support-and-wellbeing</code> (crawl/students-full, 833 pp)
2	Stand up a subdomain registry + provisioning gate	Governance	HIGH	medium	415 distinct <code>*.unimelb.edu.au</code> hosts in the contextual graph alone (cross-site-flow.csv)
3	Mandate a thin standardised student overlay on the Squiz platform you already run everywhere	Governance	HIGH	medium	Every faculty already on Squiz/Matrix (~11–13 markers/page) but no two share a top-level taxonomy
4	Fix the <code>?a=</code> query-string artifact manufacturing ~3,359 phantom 404s on live pages	Accessibility & Technical	HIGH	quick-win	32 paths 404 only with trailing <code>?a=</code> ; canonical paths live (incl. Medicine "Our Departments" ×1,143)
5	Guarantee 301s for MSD's ~8,000 deep-links into the legacy hub paths	Findability & Content	HIGH	quick-win	<code>msd→students</code> = 8,069 contextual links, ~99% on legacy <code>/your-course</code> , <code>/student-support</code> paths (cross-site-flow.csv)
6	One URL convention for the current-students entry: top-level <code>/students</code>	UX & IA	HIGH	medium	5 URL conventions at different depths; MDHS/medicine/biomed/dental bury students inside <code>/study</code>
7	Build the graduation→alumni bridge (35 completion pages, 0 alumni links)	Service Design	HIGH	quick-win	<code>students.unimelb</code> → any alumni host = 0; graduation-day page links Facebook/Instagram but not alumni (crawl/students-full)
8	Wire faculty + hub to central Careers Online at the apply-for-jobs moment	Service Design	HIGH	medium	All careers hosts get only 100 contextual links estate-wide; 5 faculties send 0 (cross-site-flow.csv)
9	Give each scholarship detail page its own <code><title></code>	Content Quality	HIGH	quick-win	2,038/2,151 (94.7%) emit identical <code>`Find a scholarship</code>
10	Fix the shared-CMS heading-skip seam at the template layer	Accessibility & Technical	HIGH	medium	678/1,523 (45%) of pages-with-headings skip a level; scholarships 96%, finearts 85% (<code>a11y_verify.py</code>)
11	Surface live 2026 handbook subjects/courses via browsable indexes + a server-rendered fallback	Findability & Content	HIGH	large	266 handbook orphans = 161 subjects + 105 courses, blank <code><title></code> , search-only (orphans.csv)
12	De-duplicate HDR candidature/thesis content mirrored across education + mdhs onto <code>gradresearch.unimelb</code>	Content Quality	HIGH	medium	33 path-tails identical (Jaccard=1.000) across two faculty CMSes; canonical owner already exists

1. UX & Information Architecture

1.1 Finish the stalled hub migration & collapse the parallel URL trees [HIGH • medium]

(merges *ia-navigation* "dual-serves 120 leaf pages" + *task-journeys* "Collapse the parallel URL trees") The canonical hub is mid-migration and runs **two parallel trees for the same content**. A query over `crawl/students-full/index.json` (833 pages) reproduces **120 leaf page-names living under 2+ top-level roots**: 75 under both `/your-course` AND `/course-admin`, 24 under both `/student-support` AND `/support-and-wellbeing`, plus a tail. The pairs are near-identical live pages with distinct fingerprints, not redirects (e.g. `/your-course/.../academic-progress` 240,879 b fp aa9d8e88 vs `/course-admin/academic-progress` 240,595 b fp f32306f0; Stop 1 served at both `/student-support/advice-and-help/stop-1` and `/support-and-wellbeing/stop-1`). The split shows up as competing inbound links: the old Stop 1 path is linked **1,648×** estate-wide vs **224×** on the new path; the old visa `advanced-standing` path **613×** vs **303×** new. The hub actually exposes 11 substantive top-level roots, not the "four" the corpus asserts (structural-findings.md Finding 23) — the lean exemplar held up to standardise faculties against is itself the most internally-inconsistent IA in the estate. **Do**: Pick one canonical root per concept (the analytics-winning URL — data suggests `/support-and-wellbeing` and `/your-course` are the newer targets), 301-redirect all 120 legacy leaves, add `rel=canonical`, and update the high-fan-in links (the 1,648 to old Stop 1, 613 to old visa). Add a redirect-coverage check so no path has live inbound on both trees. **This must land FIRST** — faculties are being told to link into a hub whose own URLs are unstable. **Acceptance test**: re-run the dual-path leaf query; target 0 leaves under 2+ roots (currently 120).

1.2 One URL convention for the current-students entry — top-level `/students` [HIGH • medium]

A "first segment containing *student*" query confirms **five conventions at different depths**: depth-0 `/students` (eng 222, law 201, arts 92, science 67, fbe 43), depth-0 `/current-students` (msd 993, ffam 47), and **depth-2+ buried inside the prospective /study funnel** (mdhs `/study/current-students` 17, medicine `/study/current-student-resources` 16, biomed `/study/current-student-information` 42, dental `/study/student-profiles` 5; education buries 107 under `/study/graduate-research/current-students`). A student's current-students home is sometimes a top-level nav item and sometimes 2+ clicks deep inside prospective content. **Do**: Mandate top-level `/students` (the modal pattern, 5 faculties) and 301-redirect `/current-students`, `/study/current-students`, `/study/current-student-information/-resources`, `/study/student-profiles` to it. Pull the MDHS/medicine/biomed/dental student sections OUT of `/study` (prospective) up to `/students` (enrolled). Redirect + nav-config task, not a content rebuild.

1.3 No two faculties share a top-level taxonomy — there is no common IA to standardise *into* [HIGH • large]

First-path-segment analysis of all 12 faculty `index.json` files: the dominant top-level section differs in **every** domain and is almost never the student section — mdhs 89% `/research` (5,387/6,072), medicine 63% `/school-structure` (2,015/3,213), ffam 51% `/about-us`, law 49% `/centres`, dental 47% `/engage`, science 43% `/engage`, msd 42% `/current-students`, biomed 30% `/departments`, fbe 23% `/newsroom`, eng 21% `/mric`, education 20% `/faces-of-education`, arts 13% `/school-of-social-and-political-sciences`. Section *count* ranges 9 (science, dental) → 54 (arts). This is the parent-architecture root cause of the five-URL-convention problem. **Do**: Do **not** try to unify parent faculty taxonomies (they serve research/news/industry, not students). Mandate a thin, consistent **student overlay** every faculty exposes identically regardless of parent IA: a fixed `/students` path, a fixed set of section labels, and a fixed "what's here vs what's on students.unimelb" block. The only lever that touches all 12 architectures without a rebuild — and it depends on the hub IA being stable first (1.1).

1.4 Research-led faculties demote the student entry below research [MEDIUM • large]

The three medical-cluster domains the corpus's 9-faculty cut omitted are the most extreme: mdhs 89% `/research` with students at `/study/current-students` (17 pp); medicine 63% `/school-structure` with students at `/study/current-student-resources` (16 pp); biomed 30% `/departments` + large `/sbs-research-groups` tail with students at `/study/current-student-information` (42 pp). These three also contribute almost nothing to the hub (medicine 21, mdhs 175, biomed 0 contextual links into students.unimelb) — confirming the "research paradox" (biggest research estates carry least student support) at the IA level. **Do**: For mdhs, medicine, biomed, the overlay mandate must promote the student entry to a stable top-level `/students` node OUT of the research/study hierarchy, plus a persistent global-nav "Current students" link. Treat these three as the priority pilot — their fragmentation cost is highest.

1.5 Navigation depth has no estate standard — content buried 9–11 levels deep [MEDIUM • medium]

URL-depth histogram over the 12 faculty `index.json` files: mean path depth 3.12 (mdhs) → 5.36 (msd) and 5.06 (medicine), max 9–11 (eng 11, law 10, biomed 10, msd 9, medicine 9). The deep tail is concentrated in marketing-led estates: MSD 397 pages at depth 6 and 888 at depth 7–8; medicine 560 at depth 6 and 716 at depth 7–8; law 507 at depth 6. `nav-depth.csv` corroborates

(msd 65% of pages ≥5 deep, medicine 59%, study.unimelb 58%). **Do:** Add a depth target (2–3 levels for task content, hard cap ~5) to the overlay template; prioritise re-nesting MSD's `/current-students` tree (993 pp) and Medicine's `/school-structure` (2,015 pp). Pair with a breadcrumb requirement so deep pages stay orientable.

2. Findability & Content Connections

2.1 MSD's 8,069 deep-links point at the legacy paths the hub is abandoning — a mass link-break waiting to fire [HIGH · quick-win]

`cross-site-flow.csv` row 1: msd → students = **8,069** contextual links, far above the next faculty (education 275, eng 224, arts 191, mdhs 175, fbe 131, law 99, science 64, biomed 0). 1,550 of 1,560 MSD pages (99%) carry the same ~8 templated deep links, each on the **legacy** top-level path (`/your-course`, `/student-support`) the hub is migrating off; only 50 MSD pages link the new `/support-and-wellbeing/stop-1`. MSD is simultaneously the best-integrated faculty and the most exposed to the unfinished migration. **Do:** (1) When 1.1 lands, guarantee permanent 301s on all legacy `/your-course` and `/student-support` URLs so MSD's ~8,000 links keep resolving; (2) MSD maintains its hub links in ONE shared include so re-pointing them is a single edit, not 1,550; (3) audit the inverse problem — most faculties barely deep-link the hub (biomed 0, science 64, law 99), so canonical task pages are near-invisible from most faculty estates (see 4.x service model).

2.2 Surface live 2026 handbook subjects/courses via browsable indexes + give them a `<title>` [HIGH · medium]

(merges findability "browsable indexes" + lifecycle "266 orphaned subject/course pages") `orphans.csv`: 266 of 289 estate orphans (92%) are handbook pages — **161** `/2026/subjects/` + **105** `/2026/courses/`, every one a current-year node, all with a **blank `<title>`**, reachable only via the handbook search box (the estate's #1 hub is a handbook `/search` URL with 5,529 contextual inbound). Inbound study links land on course/overview pages, but the constituent subject pages a continuing student needs (to check a prerequisite or plan next semester) are dead-ends. **Caveat:** per the data-integrity finding, many of these 266 are Incapsula bot-block shells — confirm via a JS-enabled re-crawl, but the underlying *browse-reachability* gap is real (the handbook is a client-rendered SPA with no crawlable index). **Do:** Generate crawlable A–Z / by-discipline / by-level index pages for the current year (e.g. a `/2026/subjects` index linking every subject and course node) and link them from the handbook home and each faculty's "courses" landing page; add `<title>` tags to the 2026 subject/course templates. Fix internal cross-linking so every subject is reachable from its parent course/major and from sibling subjects (prereq/co-req back-links), replacing JS-only subject widgets with real `<a>` links. Worklist: the 266 handbook rows of `orphans.csv`.

2.3 Fix the `?a=` artifact manufacturing phantom 404s on live pages [HIGH · quick-win]

The root cause behind the estate's apparent broken-link mass. 32 paths return "Page not found" **only** when a trailing `?a=` is appended (medicine `/school-structure?a=`, msd `/maker-spaces?a=`, biomed `/departments?a=`) — but the canonical paths are **live** (medicine `/school-structure` is a working 187KB "Departments at Melbourne Medical School"). Because URL normalisation strips the query, the `?a=` 404 twin merges with the canonical path and tags it dead. Of 4,316 contextual links that appear to hit 404s, **3,359 (78%) actually point at 30 destinations whose canonical URL is live** and 404 only via `?a=` — including the 1,143 in-content "Our Departments" links on Medicine. `hubs.csv` lists `medicine/school-structure?a=` as the #2 estate "hub" (1,146) purely from this artifact. **Do:** Find the template/component emitting the trailing `?a=` (an unresolved anchor/analytics token) and remove it, or add a server rule stripping an empty `a` param before routing. Single fix clears ~3,359 phantom-broken links and collapses ~32 phantom-404 "hubs" back into live pages. No page restoration needed.

2.4 Sweep the ~414 genuinely-dead destinations (957 links) — after the `?a=` fix [MEDIUM · medium]

After separating the `?a=` artifact, the residual genuinely-dead surface is **957 contextual links → 419 destinations** (only 5 synthetic `/404` endpoints; ~414 real gaps, thin long tail — top is `msd/404` itself at 67, then ≤8 each). Corpus-wide 493 error-titled pages (title scan of 27,302 pages; `dead-pages.csv` lists 499), concentrated in the MDHS cluster (biomedical 153, msd 74, fbe 72, mdhs 53, medicine 51). A few are live failures the CMS still links to: law `mls-staff-listing-records/.../judith-grbich` (404), `.../dudi-rulliadi` (502), mdhs `diversity-and-inclusion/.../health-and-wellbeing` (502), research `commercialisation/researchers/our-entrepreneurial-ecosystem` (502). **Do:** Run the `?a=` fix first; then a one-off sweep over the ~414 dead destinations — restore or 301 each, fix the referring template. Prioritise by faculty CMS (long tail, not a top-20). Stand up a recurring broken-link report keyed on `title='Page not found'`, **reported against canonical URLs** so `?a=` twins don't re-inflate it.

2.5 Stop treating search-result URLs as canonical destinations [HIGH • large]

The estate's #1 contextual hub is a paginated handbook **search** URL (/search?page=27&types[]=course&year=2026 , 5,529 inbound). **13,435 contextual links point INTO search/results/listing pages**, and 6,040 crawled pages ARE listing pages (~28% of true content). Other faculty "hubs" are frozen search states: mdhs /visit/research-library "Search" (929), finearts news-topics query (881), fbe /newsroom with a 7-param query string (853). **Do:** Establish canonical content pages for high-traffic intents (course discovery, news-by-topic, scholarships listing) and link to **those**, not to a frozen search state. Add rel=canonical on results pages pointing at a clean landing page; strip tracking/filter params from outbound template links; governance rule: in-content links must target a stable content URL, never a paginated/filtered search result.

2.6 Non-descriptive anchor text on the Handbook — the biggest single accessibility + findability win for the effort [MEDIUM • quick-win]

(merges findability "non-descriptive link labels" + governance "link-quality standards" for the handbook case) anchor-text.csv for the handbook: **"View detailed information in the Handbook" ×2,909**, "Handbook" 651, "University Handbook" 217, raw URL 108, "Arrow-rightHandbook link" 102, "here" 27. Almost none name the destination — a correctly-named anchor like LAWS50023 Legal Method and Reasoning appears only **16×**. Same pattern on other key destinations: findanexpert linked **267×** as **"(no text)"** (vs 204 "Find an Expert"); www.unimelb 496× "(no text)". Screen-reader users get a list of identical "Handbook"/"here" links (WCAG 2.4.4). **Do:** Estate content rule: links to the Handbook must use the subject/course name as anchor text. This is a find-and-replace-scale CMS template fix on the ~2,900 templated "View detailed information in the Handbook" instances.

2.7 Fix non-descriptive in-content link labels across the estate [MEDIUM • medium]

Measured strictly over the estate's own contextual graph (chrome-stripped, unimelb-host): **~3,300 contextual links carry vague/missing labels** — 1,805 empty/no-text, 790 bare-URL-as-text, plus ~700 generic ("read more" 833, "here" 403, "find out more" 299). (Correction: the prior 30,722/13,794-empty figure conflated external links — 11,989 of the empty-text links point to non-unimelb social/partner icons, a separate external-link-accessibility pass.) **Do:** Fix the ~1,805 empty **internal** links first (likely icon/image links missing an accessible name — add aria-label/visible text), convert the ~790 bare-URL anchors to human labels, replace generic "learn more/read more/here" with the destination title. Add a content-authoring lint rule blocking "click here"/"read more"/empty link text. Start with findanexpert and www.unimelb. Treat the ~12k empty external icon links as a footer/icon-component accessibility pass.

2.8 Retire stale duplicate ...2 school pages on mdhs [MEDIUM • quick-win]

All 6 mdhs orphans are duplicate ...2 -suffixed school landing pages — /our-organisation/melbourne-medical-school2 , /melbourne-school-of-health-sciences2 , /melbourne-dental-school2 , /melbourne-school-of-psychological-sciences2 , /melbourne-school-of-population-and-global-health2 , /school-of-biomedical-sciences2 — each a full school homepage with ZERO contextual inbound, diluting relevance against the canonical pages. **Do:** 301-redirect each ...2 page to the canonical (non-suffixed) school page and unpublish the duplicate; or if the ...2 page is the intended current version, swap and redirect. Removes 6 orphaned duplicates and search-result confusion.

2.9 Add contextual inbound links to student-critical pages that sit search-only [MEDIUM • quick-win]

orphans.csv (non-handbook scan): the international-student visa page (students.unimelb.edu.au/support-and-wellbeing/international-student-support/visas/updating-your-visa-status , "Keep your personal and visa details up to date") is a true orphan — exactly one inbound link estate-wide, and it is nav chrome. A **compliance-critical** task page reachable essentially only via search. Other notable single-domain orphans of student value: fbe/contact, law/contact, law/about/mls-video-gallery, finearts-music/faqs and /showcase, medicine/medical-history-museum. **Do:** Add in-content links to the visa-status page from its obvious parents (international student support hub, enrolment/personal-details pages, commencing-student checklist). As a governance pass, route the handful of non-handbook orphans into their natural parent sections so no compliance/support page depends solely on search.

3. Service Design, Lifecycle Seams & Ownership

3.1 Build the graduation→alumni bridge — stop handing finishing students to social media [HIGH • quick-win]

(merges lifecycle "graduate→alumni transition seam" + task-journeys "graduation→alumni bridge") Python over `crawl/students-full` (833 pp): `students.unimelb` → **any alumni host = 0**. There are **35** true graduation/completion pages (`/graduation/graduation-day`, `/testamur`, `/australian-higher-education-graduation-statement-ahags`, 16 ceremony-schedule pages, the Stop 1 graduation category page). The literal graduation-day page links out to Facebook (2), Instagram (2), LinkedIn (1) — but not to the alumni community, benefits, or `mentoring.unimelb.edu.au` (a live host, 59 references estate-wide, never linked from a graduation page). Estate-wide only 97 contextual links reach alumni hosts, and those are event RSVPs ("Register to attend" 13, "Book now" 10) — never a lifecycle handoff. The only faculty→alumni links that exist are **49 to a giving/donation "match" form**. "You are now a graduate" has no bridge to alumni identity. (Correction: earlier "83 pages" double-counted; true count is 35; the 0-alumni-links result is unchanged.) **Do:** Add a standard "After you graduate / Your next chapter" block to the ~10 highest-traffic graduation pages (both URL-tree copies) linking to: alumni community sign-up (`www.unimelb.edu.au/alumni`), alumni benefits, and the alumni mentoring platform. Replace the donation "match" form as the primary alumni touchpoint with a relationship/onboarding link (keep giving secondary). Mirror careers continuity: link Careers Online → alumni careers (today only 1 such link exists). Content insertion only.

3.2 Wire faculty + hub to central Careers Online at the apply-for-jobs moment [HIGH • medium]

`cross-site-flow.csv`: total contextual links to careers hosts (`careersonline` + `careers.unimelb`) estate-wide = **100** — fewer even than alumni. Per-faculty: dental/education/mdhs/medicine/science = **0**; biomed/law/msd = 1; arts = 10. MSD sends 8,069 links to `students.unimelb` but exactly **1** to careers. Even the central hub sends only **22** links to `careersonline` despite hosting a 36-page careers section. Engineering (15+3) and FBE (12+4+2) are the only faculties that wire it properly. **Do:** Treat `careersonline` (the transactional jobs/appointments portal) as a first-class lifecycle destination. (a) Add a contextual "Find graduate jobs & book a careers adviser → Careers Online" CTA to every faculty "current students" and course-resources landing page (template change × 13 faculties); (b) from the central careers section, deep-link "Find a job" / "Get advice" straight into the matching Careers Online module. Target: lift faculty→careers to a floor of ~18 per faculty (the eng/FBE pattern).

3.3 Resolve the HDR-support paradox at the seam — route research-intensive faculties to the Graduate Research Hub [MEDIUM • quick-win]

`cross-site-flow.csv` (faculty→gradresearch): Education **1,972** and MDHS **1,882** dominate, but research-heavy faculties barely connect — MSD **2**, biomed **3**, ffa 7, medicine 16, science 18, eng 19, fbe 20. The faculties with the most research output give their HDR candidates almost no contextual path to the central hub where the candidature lifecycle (supervision, milestones, Study Away, thesis prep) lives. `anchor-text.csv` shows gradresearch is described with real lifecycle tasks ("Working with my supervisors" 39, "Preparation of Graduate Research Theses" 40). **Do:** Add a standard "Current graduate researchers → Graduate Research Hub" contextual block (supervision, milestones, thesis-prep) to the research/HDR landing pages of MSD, Biomedical Sciences, Medicine, Science, Engineering and FBE. Quick template insertion.

3.4 Replicate the MSD central-hub model — 12 faculties under-route students vs MSD's 8,069 [MEDIUM • large]

(merges lifecycle "Replicate the MSD model" + task-journeys "Promote MSD's sidebar to a shared component") `cross-site-flow.csv`: `msd`→students = **8,069** (the largest edge in the estate) — MSD fully delegates its current-student experience to the central hub via a self-built templated sidebar (uniform ~611-count anchor block: "Advanced standing" 613, "Course advice" 611, "Leave of absence" 611, "Special consideration" 610, "Stop 1" 660...). Every other faculty is an order of magnitude lower (education 275 ... medicine 21, dental 6). One faculty solved the wiring problem locally; the other eleven didn't — which is exactly why Stop 1 routing is so uneven. **Do:** Adopt MSD's current-student link block as the **canonical shared "manage your course" navigation component** in the common faculty template, so all 12 faculties expose the same central actions (enrolment, special consideration, leave, withdrawal, study overseas, exams) and retain only faculty-specific content (placements, discipline advice, local contacts). Prioritise the worst routers — dental (6), medicine (21), mbs (26). Work is structural propagation, not copy rewriting (link-text quality on the major seams is already good).

3.5 Make Stop 1 a genuinely universal, evenly-routed front door [MEDIUM • medium]

Per-faculty links to any `students.unimelb` stop-1 path: MSD **663**, Education 54, MDHS 38, Arts 29, FEIT 29, Law 14, FBE 11, Science 11, Fine Arts 9, Biomed 7, MBS 3, Medicine **2**, Dental **1**. "Stop 1" is the single most common cross-site anchor (754) — the concept is established — but effectively only MSD surfaces it at scale. A Medicine or Dental student in difficulty has almost no faculty-page path to the central support front door. **Do:** Standardise a "Need help? → Stop 1" component in the shared faculty template (the

same pattern MSD already uses). Target parity across faculties for the core support actions; prioritise Dental (1), Medicine (2), MBS (3).

3.6 No shared scholarships-catalogue convention — hub barely reaches it [MEDIUM • medium]

`cross-site-flow.csv`: 1,865 contextual links reach `scholarships.unimelb`, scattered by faculty with no shared pattern (study 411, medicine 325, ffam 221, education 150 ...) while the 833-page hub sends only **28** and `msd` 19. `anchor-text.csv` shows inconsistent labels ("Search for scholarships" 111, "Find a scholarship" 19, bare "Scholarships" 19). Continuing-student (not just entry) scholarships are hard to reach once enrolled. **Do:** Publish one canonical "Search scholarships" deep-link (pre-filtered to current/continuing students), mandate it via a shared component in faculty templates and the `students.unimelb` finance/fees section, standardise anchor text to "Search for scholarships," and lift `students.unimelb→scholarships` from 28 toward the faculty norm (~100).

3.7 Make the prospective↔current handoff bidirectional and intentional at apply→enrol [MEDIUM • medium]

(merges lifecycle "bidirectional handoff" + task-journeys "stop bouncing students into the prospective funnel") `cross-site-flow.csv`: `study→students` = 373 (forward) but `students→study` = 125 (mostly accidental back-links). `study`'s apply-path outbound is dominated by handbook (4,391) and forms.your enquiry (1,493 — "Register for updates" 1,197); the explicit "accepted your offer, here's how to enrol" bridge to `students.unimelb/new-students/get-started` is thin (373 across 1,819 study pages). Worse, current students are **re-marketed as prospects**: `students.unimelb→forms.your` (prospective lead-capture) = **311** links; 75 faculty current-student pages link the same prospective form; current students push to study "how-to-apply" paths 95×. **Do:** On study course/"How to apply" pages add an explicit "Accepted your offer? Start here →" link to `students.unimelb/new-students/get-started` (segmented domestic/international, UG/PG). Audit the 311 `students→forms.your` and 75 `faculty→forms.your` links and replace prospective lead-capture CTAs on current-student pages with current-student equivalents (Stop 1 enquiry / `ask.unimelb` / the relevant `my.unimelb` action). Where a current page genuinely needs an apply action (advanced standing), point to the authenticated current-student path, not the marketing funnel.

3.8 Give faculty placement/WIL pages real deep-links into the careers + WIL spine [MEDIUM • medium]

211 faculty placement/WIL pages emit 347 links to `students.unimelb`, but **201 (57%) go to the bare homepage** — generic, not WIL/careers content. Only a handful reach actionable targets (`Stop 1` ×18, `/careers` ×13, `working-on-a-student-visa` ×7). `careersonline` gets just **11** inbound from all 211 placement pages combined. Placements/WIL is the most current-only, faculty-owned topic in the corpus, yet the journey dead-ends at the homepage. **Do:** Replace bare-homepage links with deep links to the central careers service and Careers Online plus WIL-relevant central pages (`working-on-a-student-visa`, insurance/safety). Establish a single central WIL hub page faculties point at, so the 211 pages share one connective target.

3.9 Unify counselling/wellbeing entry across students.unimelb and services.unimelb [MEDIUM • quick-win]

`Faculty→counselling/health` links split across two domains for the same service: the `students.unimelb` support-and-wellbeing tree AND `services.unimelb/counsel` (51), `/health` (22), `/counsel/home` (13), `/counsel/emergency-and-crisis-support` (6) — 266 `services.unimelb` `counsel/health` links from faculties. A student seeking counselling can land on either, each describing the same provider differently — for the highest-stakes task. **Do:** Designate one canonical counselling/wellbeing landing (recommend the `students.unimelb` support-and-wellbeing tree as the student-facing front, with `services.unimelb/counsel` as the operational/booking detail it links INTO). Make every faculty and central counselling link resolve to that one page. Link-target cleanup, not new content.

3.10 Upgrade alumni links from event-RSVP chrome to a named lifecycle relationship [LOW • quick-win]

`anchor-text.csv`: the 97 contextual alumni links are almost entirely transactional event RSVPs ("Register to attend" 13, "Book now" 10) or undated newsletter month labels; only a handful name an alumni relationship ("University Alumni" 3, "Chancellor's Circle" 1). The in-content alumni presence is event noise, not a benefits/identity handoff. **Do:** Where alumni links appear in faculty content (`law's` 41 `secure.alumni`, `education's` 10 `comms.alumni`), standardise anchor text and destination to a single "Join the alumni community / alumni benefits" page rather than ad-hoc event registrations. Pair with 3.1 so the relationship is introduced at completion.

4. Content Quality & Duplication

4.1 De-duplicate HDR candidature/thesis content mirrored across education + mdhs onto

`gradresearch.unimelb` [HIGH • medium]

8-word-shingle Jaccard over `crawl/ page.html : education.unimelb.edu.au/study/graduate-research/current-students/<tail>` and `mdhs.unimelb.edu.au/research/research-training/current-students/<tail>` share **33 path-tails with IDENTICAL visible content** — every sampled pair scored **Jaccard=1.000** with matching word counts (confirmation 3,072w both, my-thesis-in-the-library 3,601w both, unsatisfactory-progress 3,163w both, submitting-my-thesis 2,189w both, library 1,928w both...). The canonical owner already exists: `education→gradresearch` 1,972 + `mdhs→gradresearch` 1,882 = **3,854 contextual links**. Identical HDR policy is maintained in 2 faculty copies on top of the intended central hub — every edit must be made 2–3× and will drift. **Do:** Make `gradresearch.unimelb.edu.au` the single source of truth. Replace each of the 33 mirrored education+mdhs pages with a 301 (or a thin faculty stub linking to gradresearch), keeping only genuinely faculty-specific GR pages (25 education-only + 37 mdhs-only tails). Collapses a 3-way maintenance burden to 1.

4.2 FBE and MBS double-host the entire BCom careers tree (59 near-identical pages) [HIGH • medium]

The two domains run parallel careers trees under different prefixes — FBE at `fbe.unimelb.edu.au/bcom/career/<tail>`, MBS at `mbs.unimelb.edu.au/career/<tail>` — sharing **59 identical tails** (apply-your-learning/internship/faqs, expertise/case-competition, the whole fbe-student-calendar tree...). 8 of 10 sampled shared tails scored Jaccard ≥0.6 (clubs-and-societies 0.95, internship/workhub 0.86). The mirroring is literal: MBS hosts **FBE-branded** pages — `mbs.unimelb.edu.au/career/leaders/student-ambassador-program/meet-our-fbe-student-ambassadors`, `.../nikita-jap`, `.../nishant-sahoo` (35 MBS `/career/` URLs contain "fbe") — signalling a copy-paste site build. **Do:** Pick one owner (`fbe.unimelb.edu.au` as the faculty home) and 301-redirect the 59 duplicated career tails on MBS to it. Immediate quick fix: repoint the cross-branded URLs (MBS serving "meet-our-fbe-student-ambassadors" and the fbe-student-calendar).

4.3 "Employability in X" is a 7-faculty template clone [MEDIUM • medium]

All seven faculty landing pages titled "Employability in ..." (biomed/MDHS, msd, education, arts, ffam, eng, science) share a consistent scaffold reworded per discipline — pairwise 8-shingle Jaccard 0.28–0.52 (highest biomed↔arts 0.52), word counts 1,007–1,328. Structural duplication maintained 7×. **Do:** Replace with ONE shared central "what employability means at Melbourne" component (owned by `students.unimelb` / central Careers) injected into each faculty page, leaving a thin per-faculty block linking to that discipline's internship subjects + mentoring program. Cuts 7 drifting pages to 1 component + 7 stubs.

4.4 ~5,200 MDHS + ~2,100 scholarships pages are faceted-search permutations — 24% of all crawled URLs [MEDIUM • medium]

8,854 of 37,162 crawled URLs (23.8%) carry a `?query`. Two endpoints dominate: `mdhs.unimelb.edu.au/research/research-library/_nocache` exploded into **5,182** facet combinations (86% of all crawled MDHS URLs) and `scholarships.unimelb.edu.au/search` into **2,112** (95% of all scholarships URLs). Many render as thin "Search" shells or 502s. These crawl/SEO traps inflate every page-count metric in this project. **Do:** Add `rel=canonical` to the base listing URL, disallow facet/parameter URLs (`_nocache`, `/search?*`) in robots.txt, and add parameter-handling in the CMS/search config so facet permutations are not emitted as crawlable links. Single biggest reducer of apparent estate size.

4.5 Same content under thousands of duplicate URL forms — trailing-slash + multi-path [LOW • quick-win]

610 distinct base pages are crawled under **9,657 raw URL forms** (9,047 redundant trailing-slash/query variants). Beyond search endpoints, genuine content duplicates: `law /students/orientation` and `.../orientation/` both resolve; the FEIT Intercultural Career Mentoring Program is reachable at **5 distinct paths**. Same content at many addresses = diluted authority and ambiguous canonical. **Do:** Enforce trailing-slash normalisation + `rel=canonical` site-wide, and de-duplicate known multi-URL programs (FEIT ICMP ×5, Professional Skills Series, Arts/FFAM examples) to one canonical path with redirects. Makes every other page-count and link-graph metric in this audit more trustworthy.

4.6 A few in-content pages serve live 404/502/Cloudflare errors; 5,400+ asset/cache URLs crawled as "pages" [LOW • quick-win]

Genuine live failures the CMS still links to: `law .../judith-grbich` (404), `.../dudi-rulliadi` (502), `mdhs .../health-and-wellbeing` (502), `research .../our-entrepreneurial-ecosystem` (502); plus 11 "Just a moment..." Cloudflare-challenge pages.

Separately, **5,412 URLs are raw asset/cache forms** (`/_recache`, `/_nocache`, `.pdf`, `.m4a`, `?external-uuid=`) crawled as 0-word "pages." Low volume, high credibility cost if a student hits a 502 from an in-content link. **Do:** HTTP-status sweep over the law `mls-staff-listing-records` and research `commercialisation` trees — restore or unpublish the 404/502 pages and remove inbound links. Configure the CMS so `/_recache`, `/_nocache`, raw asset and `?external-uuid=` URLs are not surfaced as navigable pages.

5. Governance & Standards

5.1 Stand up a subdomain registry + provisioning gate [HIGH • medium]

415 distinct `*.unimelb.edu.au` destination hosts exist in the contextual link graph alone (`tail -n+2 cross-site-flow.csv | cut -d, -f2 | grep 'unimelb.edu.au$' | grep -v ezp.lib | sort -u | wc -l = 415`) — an order of magnitude beyond "20+ systems." The sprawl is self-similar: 8 Fine Arts cohort-year subdomains (`art2021–2025`, `designproduction2022–2024`), one-lab-one-subdomain research microsites (`huntlab.science`, `tigrllab.science`, `smartvillageslab.msdl`, `thirdwavelab.education...`), 19 single-purpose transaction microsites. No central inventory ties a host to an owner, a retirement date, or a reason it isn't a path on an existing site. **Do:** Create an authoritative DNS/subdomain registry (one row per host: business owner, technical owner, CMS, created date, review/retirement date, justification). Make new `*.unimelb.edu.au` issuance require sign-off against a published test ("why is this not a path under an existing site?"). Default new student-facing content to a PATH on `students.unimelb` or the faculty site. Bulk-import the 415 hosts as the seed inventory; hosts with no owner after 90 days become retirement candidates. **This is the single mechanism that prevents re-fragmentation — every other fix is one-time cleanup without it.** Retire the 8 Fine Arts cohort-year subdomains into one archived path as the proof-of-concept.

5.2 Mandate the shared CS template on the Squiz platform you already run everywhere [HIGH • medium]

(*this is the platform mechanism for the IA overlay in 1.3*) Fingerprinting raw `page.html` across 13 domains: every faculty domain returns Squiz/Matrix markers (~11–13/page) plus Google Tag Manager — the **CMS and analytics layer are already standardised and centrally controlled**. What is NOT standardised is the template/IA above it (four URL conventions, 8–17 top-level sections, no shared vocabulary). The University doesn't need a new platform — it needs to ship the template as an enforced Squiz parent template. (*Note: handbook.unimelb returns ZERO Squiz markers — the one domain not on this platform; see 6.x.*) **Do:** Publish the Education/Arts central-and-spoke pattern as a versioned, mandated "Standard Faculty Student Section" Squiz parent template: fixed URL stem `/students`, capped 2–3-level nav, mandatory "what's here vs what's on `students.unimelb`" signpost block, canonical section labels. Ships as a parent-template rollout with an adoption deadline, not a rebuild. Gate template changes through one owner so the next redesign can't re-fork the IA.

5.3 Bring the Handbook up to the estate's web-standards baseline [HIGH • medium]

On a 33-page handbook sample, `rel=canonical` is present on just **8/33 (24%)** and `og:title` on 13/33 (39%) — while every other domain sampled sits at ~100% for both (`arts` 20/20, `law` 14/14, `study` 16/16, `students` 25/25, `eng` 22/22...). The Handbook holds 266 of the estate's 289 contextual orphans (92%) vs 0–9 for every other domain. Root cause is visible: the Handbook returns ZERO Squiz markers while all 12 faculty domains return ~11–13/page — it is on a separate platform governed to a visibly lower standard than the sites that depend on it (3,380× the single biggest external dependency; 7,819 contextual inbound, 2,909 using the generic "View detailed information in the Handbook" anchor). **Do:** Treat the Handbook as in-scope for the same baseline: add `rel=canonical` and Open Graph tags to its template, fix the orphans (every subject/course reachable from a parent index, not just search), and assign it a named owner inside the same governance forum as the faculty sites — today it sits outside it.

5.4 Adopt a canonical-host rule and 301 the 28 www-vs-bare duplicate hosts [MEDIUM • quick-win]

28 sites are linked as BOTH `www.x` and `x` in the contextual graph (reproduced from `cross-site-flow.csv`): www.arts/arts, www.eng/eng, www.mdhs/mdhs, www.msdl/msdl, www.services/services, www.findanexpert/findanexpert, www.alumni/alumni, www.unimelb/unimelb, plus 19 more (`aii`, `art-museum`, `asiainstitute`, `bsc`, `cis`, `commercial`, `csdila`, `ereseach`, `gsa`, `lms`, `microbiol`, `microscopy`, `mobility`, `murrupbarak`, `ormond`, `queens`, `scc.ms`, `socialequity`, `sport`). Each splits inbound link equity and means bookmarks/anchors disagree on the "real" address. **Do:** Pick one canonical form (recommend bare host, matching `study/students/handbook`), add server-level 301s for all 28 families, plus `rel=canonical`. Add a CI/governance check flagging any new host advertised in both forms.

5.5 Consolidate the 19 single-purpose form/transaction microsites behind one forms registry [MEDIUM • large]

The link graph exposes 19 single-purpose transaction hosts — **13 on *.app.unimelb.edu.au** (specialconsideration.app, enrolmentvariations.app, studentefor.ms.app, feit-vocational-placement.app, mycounselling.app, tes.app, thesislibrary.app, booklibrary.app, slidelibrary.app, enquiry.app, ctrs.app, law.app, ffam-mcm.app) plus 6 standalone form hosts (feerefundform, casemanagementform, captureme, overseasstudyplanner, internationalstudenttravelpass, intlstudaccept). students.unimelb has no forms index — its 19 "form"-matching pages are scattered individual pages, not a discovery surface. **Do:** Publish a single authoritative forms/transactions registry at students.unimelb (task → form → owning system) and route every *.app microsite through it for discovery. Longer term, provision new student transactions as paths within a shared forms platform rather than net-new app.* subdomains; require existing ones to register an owner and retirement review. Pair with the subdomain-registry gate (5.1).

5.6 Enforce link-quality + access-boundary standards (kill empty anchors, stop staff-intranet leakage) [MEDIUM • medium]

anchor-text.csv shows no enforced standard. (1) **Accessibility/SEO:** findanexpert's most-frequent inbound anchor is "(no text)" 267x. (2) **Vocabulary drift:** careersonline has 8+ distinct anchor strings for the same function ("Express interest here", "register via Careers Online", "register via Careers online"...); the handbook's top anchor coexists with 7+ variants. (3) **Access-boundary leak:** faculty current-student estates link into the STAFF intranet staff.unimelb **838x (eng)** and **729x (msd)** (anchors "FEIT Intranet" 836, "Staff intranet" 705) — students routed to a staff system. **Do:** Add three rules to the web standard, enforced by a nightly crawl/lint check in CI (build_linkgraph.py already exists): (a) no empty/non-descriptive anchor text (block "(no text)", "here", "click here", "view more", bare titles); (b) one canonical link label per destination function; (c) student-facing pages must not deep-link the staff intranet — audit the eng (838) and msd (729) staff.unimelb links and repoint or gate them.

5.7 Annotate the handbook crawl artifact so a false "14% orphaned" doesn't reach the deck [LOW • quick-win]

1,470 of 2,134 crawled handbook pages (69%) are Imperva/Incapsula bot-block responses (body "Request unsuccessful. Incapsula incident ID...", ~885 bytes). These ARE the same 266 handbook orphans / "14% orphaned" figure in SUMMARY.md — the first orphan, .../2026/subjects/medi90100, is literally the first blocked page. Any content-quality/thinness/orphan conclusion about the handbook is a crawl artifact (1,506 such blocks estate-wide). **Do:** Annotate SUMMARY.md and orphans.csv to flag handbook entries as "crawl-blocked (Incapsula), not orphaned/thin," and re-crawl the handbook with a politeness delay / allow-listed agent before drawing handbook content conclusions. (The browse-reachability, SSR-fallback, metadata and enrol-action findings remain valid independently.)

6. Accessibility & Technical

6.1 Fix the shared-CMS heading-skip seam at the template layer [HIGH • medium]

Heading-order over rendered (non-shell) pages: **678/1,523 (45%)** of pages-with-headings skip a level — a shared Squiz template pattern, not random authoring. Per-domain: scholarships 240/250 (96%), finearts 213/250 (85%), study 201/250 (80%), mbs 70%, law 48%, fbe 41%, services 30%. The first-skip signature is template-specific (study/fbe/finearts h1→h3, law/mbs h1→h4, scholarships h2→h4). Concrete: study.unimelb.edu.au/find/courses/graduate/graduate-certificate-in-digital-engineering renders h1 then immediately h3 "Overview" (no h2). WCAG 1.3.1/2.4.10. **Do:** Fix at the design-system layer. Audit the Squiz content-block components (accordion, card, callout, section panels) that hard-code <h3> / <h4>: make the heading level a template parameter derived from nesting depth. The scholarships detail template skips h3; the course-finder template skips h2 — both need the level parameterised. One component-library change propagates to ~15 sites. Add an axe-core heading-order rule to the CMS publish/CI step.

6.2 Give each scholarship detail page its own <title> [HIGH • quick-win]

Over all 2,151 cleanly-rendered scholarships.unimelb pages, only **109 distinct <title> values; 2,038/2,151 (94.7%)** emit identical Find a scholarship | University of Melbourne. Named award pages ("Melbourne Chancellor's Scholarship", "Research Training Program Scholarship") are indistinguishable in tabs, bookmarks, history, screen-reader announcements, and collapse into one search-result entry. The largest duplicate-title group in the estate. WCAG 2.4.2. **Do:** Set <title> from the

scholarship-name field: {Scholarship name} | Scholarships | University of Melbourne (the name already renders in the body/H1). Same fix for the mdhs "Search" and study/handbook "Find a course"/"Search" result pages — append the active query/filters so result states are distinguishable.

6.3 Stop card grids from emitting multiple `<h1>` s [HIGH • medium]

Multi-h1 is template-driven and concentrated: mdhs **183/250 (73%)** and msd **88/250 (35%)** of rendered pages carry `>1 <h1>`. Worst case: `mdhs.unimelb.edu.au/research/research-library` renders the page heading AND every result-card title as `<h1>` (11 h1s on one page: "MDHS Research Project Database", "Centre for Health Equity", "Baker Department of Cardiometabolic Health Research"...). Multiple h1s break the document-outline cue screen-reader users rely on. WCAG 1.3.1. **Do:** In the research-library and MSD studio card-grid components, demote per-card titles from `<h1>` to `<h3>` (under the page's single `<h1>` + a section `<h2>`), keeping visual size via CSS. One component edit fixes the whole grid across every filtered view. Re-run axe page-has-heading-one + a custom single-h1 check in CI.

6.4 Add alt text in the Funnelback/Squiz search-result card template [MEDIUM • quick-win]

A single shared search-component defect, NOT content authoring — content domains are clean (students 0.2% imgs missing alt, arts 0%, alumni 0%, mdhs 0.5%, law 0.7%). But the scale is large: **EVERY scholarships search/listing image lacks alt — 100% (4,346/4,346)** in a 250-page sample, on ~291 search/listing URLs at ~26 thumbnails each). study /find/ shows the same (17.8% of study's imgs missing alt). One template powers both the course finder and the scholarship finder. WCAG 1.1.1. **Do:** Edit the Funnelback/Squiz result-card display template to output `alt` from the result's title/metadata (`alt="{course name}"` / `alt="{scholarship name}"`, or `alt=""` if purely decorative beside a visible title). One change clears thousands of missing-alt images across hundreds of listing URLs. No content-author action.

6.5 Add `rel="noopener"` and a visible "opens in new window" cue to external links [MEDIUM • medium]

Of 11,716 `target=_blank` links sampled across 9 domains, **8,535 (72.8%) carry no `rel=noopener/noreferrer`**. Rate varies by template: scholarships 98.0%, services 99.6%, fbe 98.4%, arts 98.4%, mdhs 98.5%, alumni 90.7%, students 80.1% — but study 31.2% and law 47.3% are markedly better. No programmatic "opens in new window" indication. WCAG 3.2.5 (unannounced context change) + reverse-tabnabbing hygiene. **Do:** At the link-rendering/template level, auto-inject `rel="noopener noreferrer"` on any anchor with `target=_blank`, and append a visually-hidden " (opens in a new window)" span (or an aria-labelled external-link icon). Priority: the ~98%-rate templates (scholarships/services/fbe/arts/mdhs). Apply as a global template/JS post-process to cover all 20 domains; add an axe check for new `target=_blank` links without `rel`.

6.6 Treat handbook subject/course pages as JS-only — provide a server-rendered fallback [MEDIUM • large]

69% of handbook captures with HTML (1,470/2,134) returned an 887-byte Imperva/Incapsula challenge with NO `<title>` / `<h1>` / `<main>` /lang/meta; for subject/course URLs specifically it is 1,332/1,718 (78%) shells. The real handbook pages are client-rendered (SPA, no SSR). Any client without full JS (some assistive setups, link previews, search engines, the University's own crawlers) sees an empty document — and the handbook is the estate's single most-linked destination (study→handbook = 4,391). **Do:** Confirm via a JS-enabled audit (axe DevTools on the live page) that rendered pages set title/h1/landmarks correctly; the SSR/WAF gap is the actionable item. Add SSR or a static HTML fallback (title, h1=subject name, meta description, core facts) for `/subjects/*` and `/courses/*`, and ensure the bot-protection does not serve the bare challenge to legitimate non-JS user-agents. Also protects link-preview and SEO equity for thousands of inbound study/faculty links.

6.7 Wire an "Enrol in this subject" action onto every handbook subject & course page [HIGH • medium]

`crawl/handbook-...` index lists 3,000 pages (1,637 subject + 818 course). Subject pages link to **my.unimelb** (the SIS where enrolment happens) **0%** of the time and to a real enrolment how-to on students.unimelb **0%**. The only students.unimelb links on subject pages (154 each) go to fees and conduct boilerplate. Top outbound hosts from subject pages are the handbook itself, www/about chrome, then Facebook/Twitter/LinkedIn/Instagram (154 each) — all ahead of any enrolment action. The catalogue is a read-only dead-end: a student can find a subject but cannot act on it. **Do:** Add a persistent templated action block keyed off the subject/course code already in the URL: (1) "How to enrol in subjects" → `students.unimelb.edu.au/your-course/manage-your-course/subject-enrolment/enrolling-in-subjects`, (2) "Add this subject in my.unimelb" → the my.unimelb enrolment entry point. Because the code is in every URL, this is a single template change, not ~1,700 page edits. Closes the find→act gap for the highest-traffic enrolment task.

6.8 Replace generic "Show more"/"Find out more" link text and fix `href="#"` placeholders [LOW • quick-win]

Generic phrases are low-volume (~0.21% of 39,941 anchors on rendered students.unimelb pages): "find out more" ×26, "learn more" ×24, "show more" ×13. **23 instances of generic-text links point at `href="#"`** (non-functional without JS) — e.g. `students.unimelb.edu.au/support-and-wellbeing/student-services-workshops-and-sessions`. WCAG 2.4.4. **Do:** Where "Show more" toggles content, use a real `<button aria-expanded>` not `<a href="#">` (fixes all 23). Where the link is real, give it a self-describing label or `aria-label`. Targeted content cleanup on students/study, not a template change.

6.9 Repair the analytics privacy-consent radios (`<label>` commented out, `{{opt_in}}` never renders) [LOW • quick-win]

Exactly **816 students.unimelb pages** carry the analytics opt-in radio, and all **816/816** ship it broken: `<label for="privacy_pref_optin">` is HTML-commented out and the Mustache token `{{opt_in}}` was never substituted, so the inputs have no programmatic label. The radios are `display:none` so live impact is low, but it is a latent WCAG 1.3.1/4.1.2 bug — and it was the source of the apparent "unlabelled inputs" signal (there are no unlabelled text/email/search fields, so the visible form story is fine). **Do:** In the consent component, uncomment the `<label for>` elements and populate `{{opt_in}} / {{opt_out}}` at render (or hard-code the visible label). Trivial fix; eliminates a dead control and the false signal across all 816 pages.

7. Performance & Mobile

*Round-2 dimension (round 1 omitted both). Mobile **setup** is fine — viewport meta is present on 600/600 sampled pages; the problem is payload weight and render cost, which hits students on phones/limited data hardest.*

Multi-MB pages from base64-inlined hero images crush mobile/limited-data students [HIGH • medium]

Evidence: First-hand HTML analysis of all 154 pages >500KB in /crawl found ~9 pages shipping >100KB of base64-encoded images embedded directly in the HTML document (not as cacheable image files): `finearts-music.unimelb.edu.au/about-us/news/meet-nicole-goode-honours-visual-art-graduate` = 2,292KB total, 2,071KB is a SINGLE inlined photo; `mdhs.unimelb.edu.au/diversity-and-inclusion/initiatives/dei-conference` = 2,704KB (2,501KB base64); `msd.unimelb.edu.au/thus/news-events/urban-density-and-access-to-social-services` = 1,779KB (1,617KB); `mdhs .../mdhs-diversity-and-inclusion-grants` = 1,220KB (1,054KB); `law.unimelb.edu.au/news/MLS/from-education-to-precision-use` = 1,145KB (976KB); `biomedicalsciences .../cmr/about-us/contact-us` = 734KB (570KB). `page-weight.csv` confirms these as the per-domain max outliers (finearts max 2,292KB, medicine 2,516KB, msd 1,779KB). Base64 inlining defeats browser caching, lazy-loading, responsive srcset, and CDN image optimization, and inflates the HTML ~33% (encoding overhead) — every byte is re-downloaded on every visit. On a 3G/limited-data plan a 2MB news article is a ~10–20s wait and a real data cost.

Do: Add a CMS publish-time guard (and a crawl-based linter over freshness-signals tooling) that blocks/flags any inline data:image payload over ~10KB; convert offending images to served files with width-appropriate srcset, `loading="lazy"`, and modern formats (WebP/AVIF). Run a one-off sweep to de-inline the ~9 known >100KB offenders. Set a hard page-weight budget (e.g. 500KB HTML) enforced in CI so editors cannot paste large base64 blobs from Word/design tools again.

Cookiebot external-link-arrow icon duplicated ~1,925× per page inflates staff pages to 5MB [HIGH • quick-win]

Evidence: `law.unimelb.edu.au` staff-listing pages (e.g. `/mls-staff-listing-records/.../luis-eslava/luis-eslava` and 3 sibling profiles) are 5,007–5,056KB each — but contain almost NO real content. Analysis shows 1,926 identical tags, all carrying the SAME ~400-byte base64 PNG (class `CybotExternalLinkArrow`, the Cookiebot 'opens in new window' arrow), i.e. the same string repeated 1,926 times instead of referenced once. `mdhs.unimelb.edu.au/research/research-training/supervisors/admissionsscholarships/times-higher-education-world-university-rankings` = 4,730KB with 1,924 copies. These are the absolute max outliers in `page-weight.csv` (law max 5,056KB, mdhs max 4,730KB). `gzip` masks but does not fix it (the law page still parses 5MB of DOM = 1,938 nodes the phone must build). Root cause is the Cookiebot consent script re-decorating every external link per render combined with these pages embedding an external page's link soup.

Do: Quick config fix: configure Cookiebot to reference the external-link-arrow as a single shared sprite/CSS background (or disable its auto-icon and use a `CSS ::after`), eliminating ~770KB+ of duplicated base64 per affected page instantly. Audit the law

'mls-staff-listing-records' template — it appears to embed full external-site markup; strip to the actual profile content. This is the cheapest high-impact win in the estate: a settings change drops 4 law pages from ~5MB to well under 200KB.

Student course template ships ~327KB of document weight before a single external file loads [HIGH · medium]

Evidence: The study.unimelb /find/courses/ template is the highest-traffic student-facing page type (course structure/entry/how-to-apply). Sampling 25 such pages: median HTML 327KB, of which ~109KB inline SVG, ~79KB inline JS, ~43KB inline CSS — i.e. ~70%+ of the document is inlined assets, not content. The single biggest inline SVG is a 106KB icon sprite sheet (187 icons) embedded in EVERY course page though most icons are never shown. study.unimelb median across all pages is 304KB (page-weight.csv), the heaviest domain by median in the estate; max 1,132KB (master-of-environment/structure). Because it is inlined per-page it cannot be cached across the multi-page course journey (structure → entry-requirements → fees → how-to-apply), so a prospective/continuing student re-downloads the same 106KB sprite + JS + CSS on every step.

Do: Externalize the shared icon sprite, base CSS, and base JS into cacheable files referenced (not inlined) so they download once for the whole course journey; keep only above-the-fold critical CSS inline. Tree-shake the icon sprite to icons actually used on course pages. Target: cut the per-page course document from ~327KB to <120KB, with the heavy shared assets cached after the first page. This directly speeds the most common student task flow on mobile.

~18–21 render-blocking external scripts (mostly third-party tags) on key student pages [MEDIUM · medium]

Evidence: Sampling 20 pages/domain for external resource counts: study.unimelb course pages load a median of 10 external stylesheets + 21 external